Chapter 4 - Traveling Waves - Notes

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1 Traveling Waves

1.1 Traveling Waves

Waves represented by functions/solutions in the form $u(x,t) = f(x - ct)$ are called *traveling waves*, where f describes the shape of the wave and $|c|$ is the speed of the wave.

If $c < 0$ then the wave moves in the negative x direction. If $c > 0$ then the wave moves in the positive x direction.

It is assumed that f is not constant and c is not 0 to represent movement through a medium.

Example: $e^{-(x-5t)^2}$ is moving in the positive x direction with speed 5.

A **traveling wave solution** of a PDE is a solution of the differential equation which has the form $u(x,t) = f(x - ct)$. You can begin finding traveling wave solutions by assuming $u(x,t) = f(x - ct)$ and then determining which functions f and constants c yield a solution to the differential equation.

Example: We want to find a traveling wave solution to the wave equation

$$u_{tt} = a u_{xx}, \quad a > 0 \text{ constant}$$

Assuming that $u(x,t) = f(x - ct)$, we can use the chain rule.

$$\begin{aligned} u_t(x,t) &= f'(x-ct)(x-ct)_t = -cf'(x-ct) \\ u_x(x,t) &= f'(x-ct)(x-ct)_x = f'(x-ct) \end{aligned}$$

And we can use the chain rule again.

$$\begin{aligned} u_{tt}(x,t) &= f''(x-ct)(x-ct)_t = c^2 f''(x-ct) \\ u_{xx}(x,t) &= f''(x-ct)(x-ct)_x = f''(x-ct) \end{aligned}$$

We can then plug our results back into the wave equation and re-arrange so it looks nice.

$$\begin{aligned} u_{tt} &= a u_{xx} \\ c^2 f''(x-ct) &= a f''(x-ct) \\ c^2 f''(x-ct) - a f''(x-ct) &= 0 \\ f''(x-ct)(c^2 - a) &= 0 \end{aligned}$$

When can then sub in $z = x - ct$

$$f''(z)(c^2 - a) = 0$$

In order for the above to be satisfied, $c^2 = a$ and f must be twice differentiable, or $f'' = 0$ (meaning f must be some linear function $f(z) = A + Bz$). Example solutions:

$$\begin{aligned} u(x,t) &= f(x - \sqrt{a}t) \\ u(x,t) &= f(x + \sqrt{a}t) \\ u(x,t) &= sin(x - \sqrt{a}t) \\ u(x,t) &= A + B(x - ct) \end{aligned}$$

1.2 Wave fronts and pulses

A traveling wave represented by $u(x,y)$ is said to be a *wave front* if for any fixed t,

$$u(x,t) \to k_1 \text{ as } x \to -\infty, \quad u(x,t) \to k_2 \text{ as } x \to \infty$$

where k_1 and k_2 are constants. k_1 and k_2 do not have to be equal. If $k_1 = k_2$, then the wave front is called a *pulse*. A *pulse* temporarily changes the value of u at position x before it settles back to its original value.

A physical example could be a cold front in weather. Temperatures ahead of the disturbance can be a relatively constant k_1, and the temperatures behind the disturbance can be around k_2. The disturbance can cause a sudden drop in the temperature (sometimes more than 30 degrees F) from k_1 to k_2.

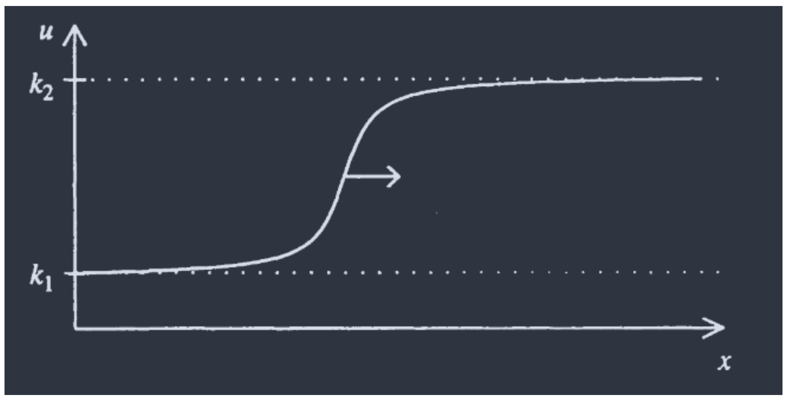

Figure 1: The profile of a wave front at time t.

1.3 Wave trains and dispersion

A traveling wave which can be written in the form

$$u(x,t) = Acos(kx - \omega t) \quad or \quad u(x,t) = Acos(kx + \omega t)$$

where $A \neq 0$, $k > 0$ and $\omega > 0$ are constants is called a *wave train*. Rewriting the $u(x,t)$ as

$$u(x,t) = Acos\left[k\left(x - \frac{\omega}{k}t\right)\right]$$

shows us that it is a traveling wave with the shape $f(z) = Acos(kz)$ and moving with the speed $c = \frac{\omega}{k}$ More generally, wave trains are represented as $u(x,t) = f(kx - \omega t)$ where $f(z)$ is a periodic function.

a) k is called the *wave number* and represents the number of cycles on this period wave that appears in a window of length 2π on the x-axis.

b) The number ω is called the circular frequency and represents the number of cycles of the wave that pass by any fixed point x on the x-axis during a time interval of 2π.

A PDE may have solutions that are wave trains but not for every possible value of k and ω.

To find which wave numbers and frequencies are permitted, one can substitute the form of a wave train such as $u(x,t) = Acos(kx - \omega t)$ into the differential equation and reduce it to a relationship between k and ω. This relationship is called a *dispersion* relation and indicates which values of k and ω may be selected in order for $u(x,t)$ to be a wave train solution.

A partial differential equation which has wave train solutions is said to be *dispersive* if waves trains of different frequencies ω propagate through the medium with different speeds.

A PDE is dispersive only if its phase velocity $v_p = \frac{\omega}{k}$ is not constant, meaning $\frac{d^2\omega}{dk^2} \neq 0$. If all wave trains travel at the same speed regardless of their ω, the PDE is not dispersive.